

Deep Dive into Behavioral Risk Factors that Might Lead to Cognitive Decline

Team Name: Deep Dive Healthcare

Xinya Li

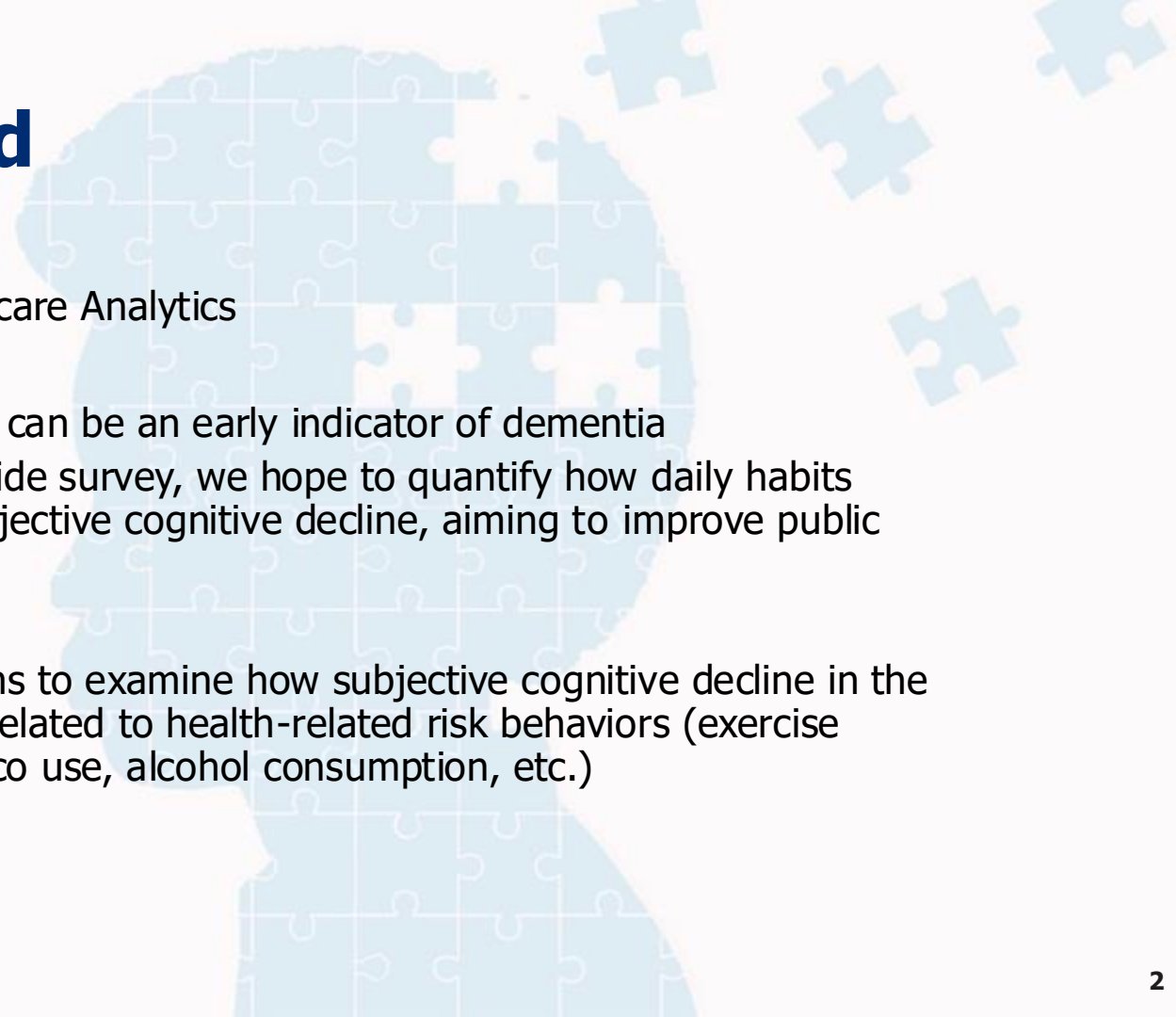
Lavanya Shankar

Zhiquan Shen

Keni Xue

Guanchen Xiao

Background



- Subject Area: Healthcare Analytics
- Motivation:
 - Cognitive decline can be an early indicator of dementia
 - Using a nation-wide survey, we hope to quantify how daily habits influence the subjective cognitive decline, aiming to improve public health.
- Proposal Research:
 - This research aims to examine how subjective cognitive decline in the United States is related to health-related risk behaviors (exercise frequency, tobacco use, alcohol consumption, etc.)

Introduction To Dataset



- Behavioral Risk Factor Surveillance System (BRFSS) is an annual, nationally representative survey in the United States
- The data is a cross-sectional phone survey (including landline telephone and cellular telephone) initiated by the CDC (Centers for Disease Control and Prevention) and conducted by state health departments.
- The 2022 BFRSS collects data in all 50 states, the District of Columbia and three US territories (Guam, Puerto Rico, and the US Virgin Islands).

Introduction To Dataset

- In the 2022 data, there are 445,132 records and 328 features in total.
- The optional cognitive decline module includes self-reported scale pertaining to challenges in daily life due to memory loss and growing confusion over the past twelve months.
- A total of more than 60,000 aged 45+ adults residing in the US completed the optional BRFSS cognitive decline module



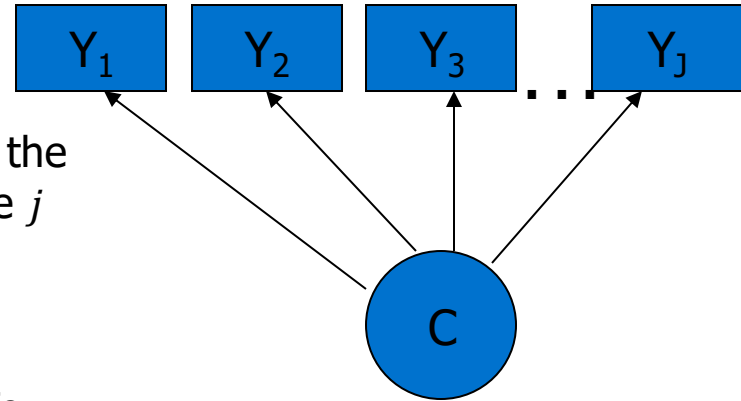
Latent Class Analysis - Motivation

- We want to extend the classification problem from binary to multi-class
- Question:
 - Among those with subjective cognitive decline, is there some unobserved hierarchy of different severities?
 - Are there specific characteristics that identify a certain level of severity?
 - Given known idea of someone with severe cognitive decline, what characteristics are important indicator?
- Answer: Latent Class Analysis!
 - To find subgroups of respondents that are similar based on CD level
 - Identify and enumerate the number of subgroups
 - Classify subjective CD into different severity levels

Latent Class Analysis – Method Review

Maximize likelihood:

- Let $\mathbf{Y}$ be the matrix of observed categorical responses for N individuals on J indicator variables, where each element y_{ij} indicates the response of individual i on indicator variable j
- Let $\boldsymbol{\pi}$ be the vector of class membership probabilities, where π_k is the probability of belonging to latent class k
- Let $\boldsymbol{\theta}_k$ represent the vector of parameters for latent class k



We want to maximize with respect to $\boldsymbol{\pi}$ and $\boldsymbol{\theta}$

$$L(\boldsymbol{\pi}, \boldsymbol{\theta}) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \times P(\mathbf{Y}_i | \boldsymbol{\theta}_k)$$

Latent Class Analysis – Model Selection

# Classes	Log Likelihood	AIC	BIC	Entropy	LMR-LRT
1	-115777	231594	231775	1.82	
2	-51626	103335	103706	0.81	$p < 0.001$
3	-48822	97769	98331	0.77	$p < 0.001$
4	-51626	103419	10417	0.81	$p = 1.000$

Latent Class Analysis – Results

		Subjective CD		
		Severe	Moderate	Mild
		13.5%	31.0%	55.5%
CDHOUSE	Always	27.6%	0.0%	0.0%
	Usually	26.4%	3.4%	0.5%
	Sometimes	39.3%	44.7%	4.7%
	Rarely	3.1%	33.2%	12.1%
	Never	3.7%	0.0%	81.5%
CDASSIST	Always	25.6%	1.4%	0.4%
	Usually	23.9%	1.4%	0.2%
	Sometimes	40.5%	34.8%	3.1%
	Rarely	3.5%	45.2%	9.8%
	Never	6.5%	18.1%	86.6%
CDSOCIAL	Always	35.6%	2.8%	0.7%
	Usually	25.8%	6.1%	0.2%
	Sometimes	28.7%	38.0%	4.2%
	Rarely	2.6%	36.4%	13.5%
	Never	7.3%	16.7%	81.4%
CDDISCUSS	Yes	74.7%	58.6%	34.8%
	No	25.2%	41.5%	65.0%

- We can see a clear hierarchy of subjective cognitive decline from mild to severe
- Those with severe cognitive decline are more likely to answer always to these questions
- We assigned labels to samples by picking the sub-class with the highest posterior probability

Individual Feature Analysis

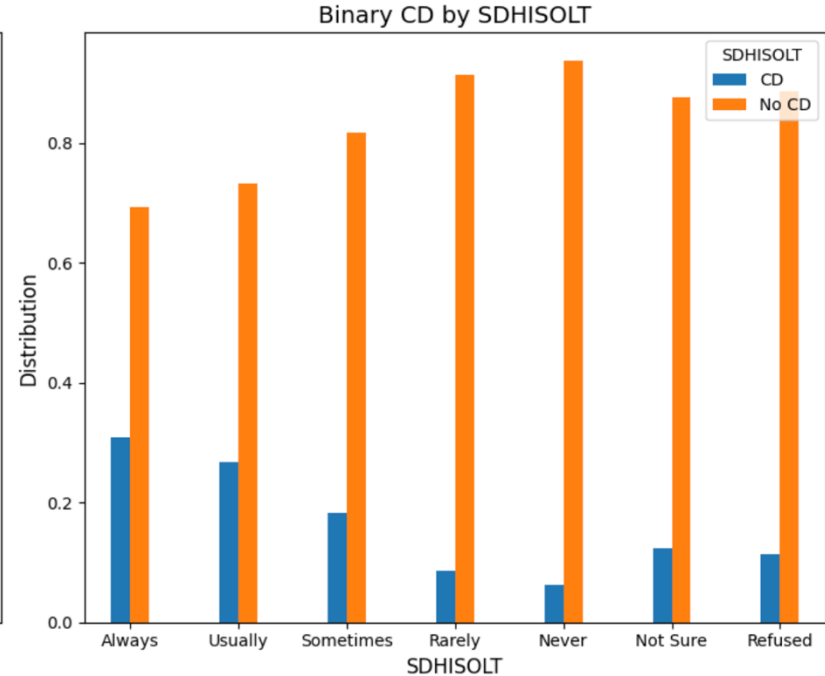
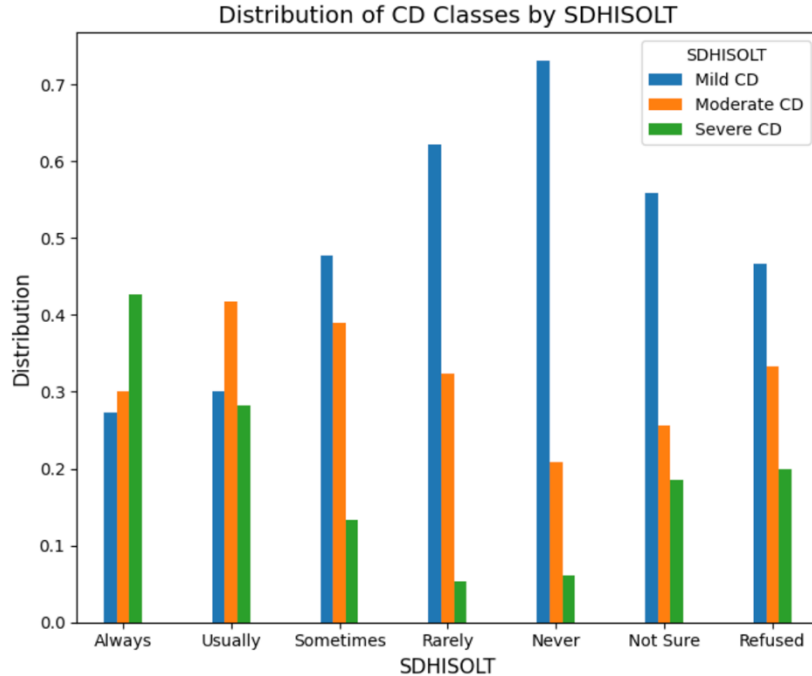
- Categorical Variables
 - Cramer'V Coefficient:
 - Cramer's V is a measure of association between two nominal (categorical) variables, providing a value between 0 and 1 that indicates the strength of their association.
 - Cramér's V is computed by taking the square root of the chi-squared statistic divided by the sample size and the minimum dimension minus 1:

$$V = \sqrt{\frac{\varphi^2}{\min(k-1, r-1)}} = \sqrt{\frac{\chi^2/n}{\min(k-1, r-1)}},$$

where:

- φ is the phi coefficient.
- χ^2 is derived from Pearson's chi-squared test
- n is the grand total of observations and
- k being the number of columns.
- r being the number of rows.

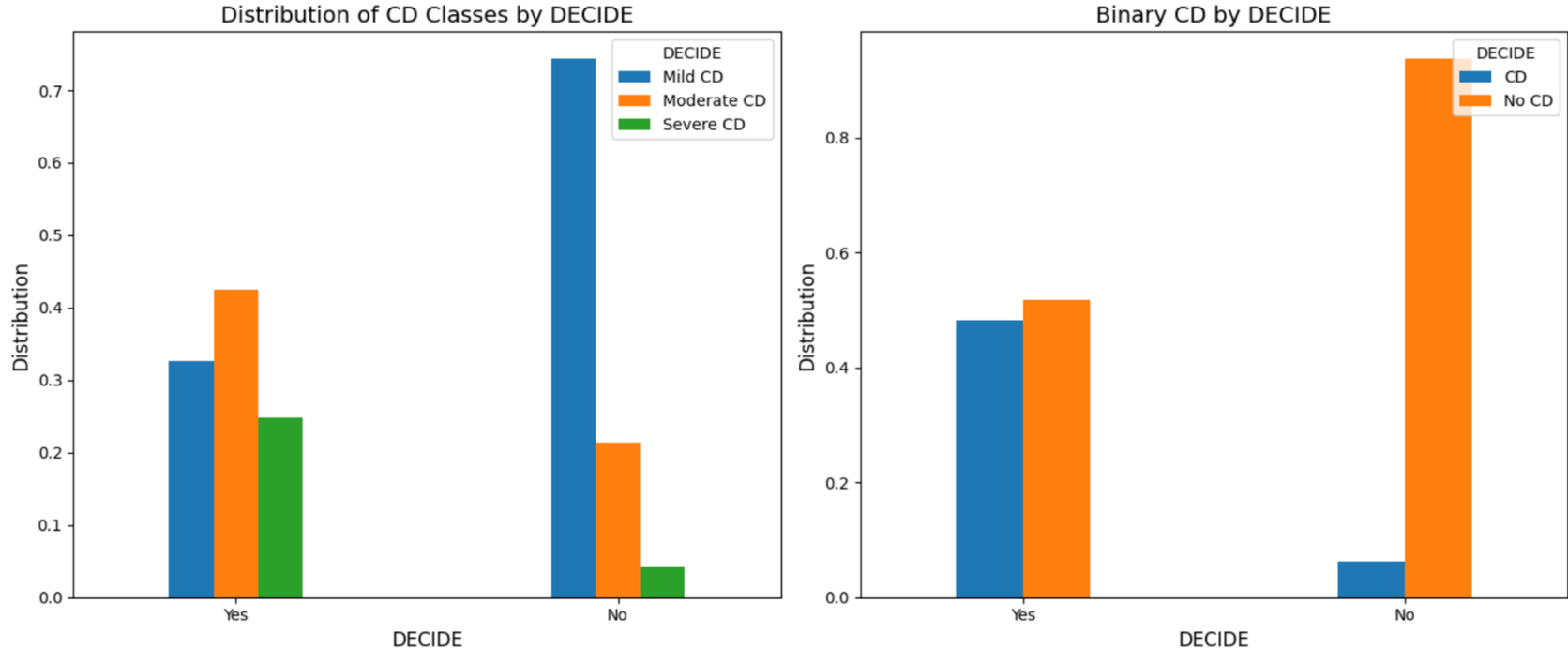
Individual Feature Analysis : Social Isolation



Cramers V for multi-class classification is: 0.2721837696519268

Cramers V for binary classification is: 0.20884010594896135

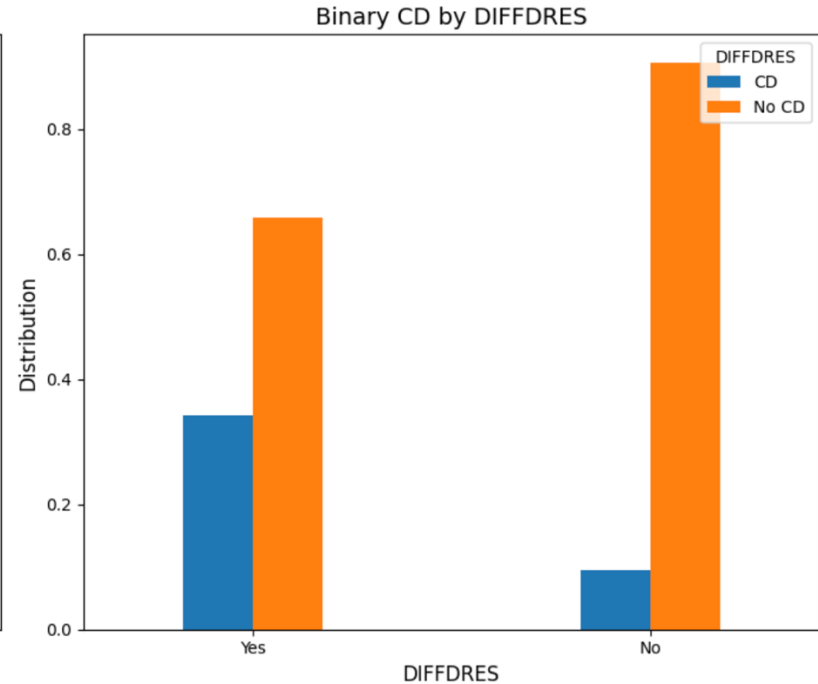
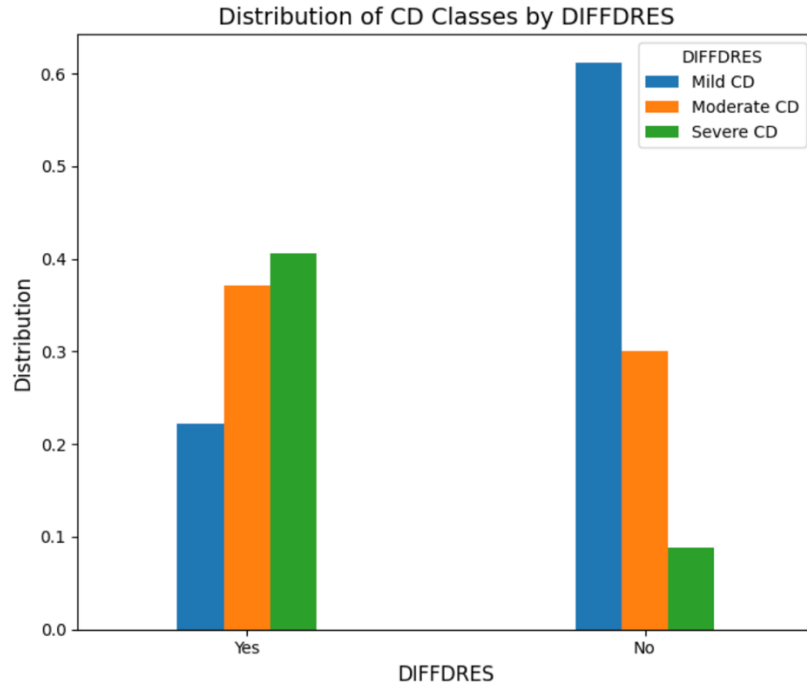
Individual Feature Analysis: Difficulty Concentrating or Remembering (Removed)



Cramers V for multi-class classification is: 0.43636442637918843

Cramers V for binary classification is: 0.4076296382456461

Individual Feature Analysis : Difficulty Dressing or Bathing



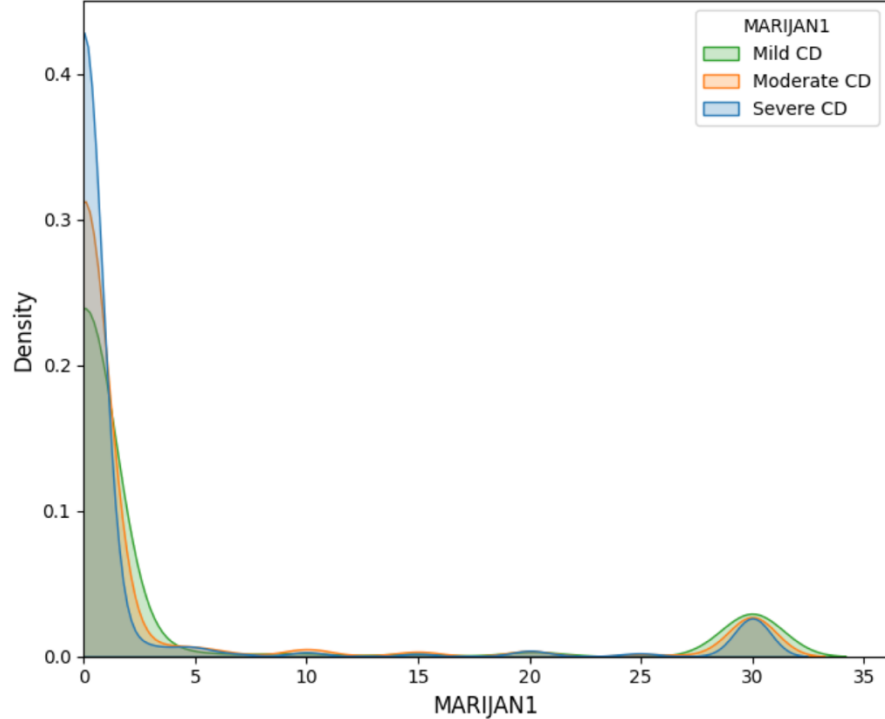
Individual Feature Analysis

- Numerical
 - ANOVA:
 - Analysis of variance (ANOVA) is a collection of statistical models and their associated estimation procedures (such as the "variation" among and between groups) used to analyze the differences among means.
 - $H_0: \mu_1 = \mu_2 = \mu_3 \dots = \mu_k$
 - H_1 : Means are not all equal.

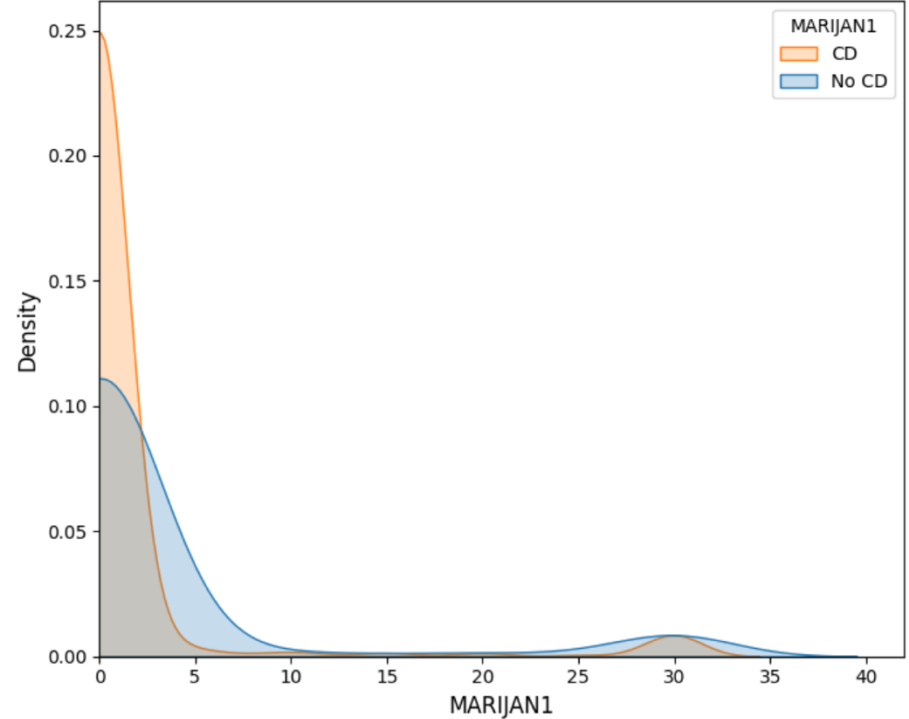
Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Value
Between Groups	$SSB = \sum n_j(\bar{X}_j - \bar{X})^2$	$df_1 = k - 1$	$MSB = SSB / (k - 1)$	$f = MSB / MSE$
Error	$SSE = \sum \sum (X - \bar{X}_j)^2$	$df_2 = N - k$	$MSE = SSE / (N - k)$	
Total	$SST = SSB + SSE$	$df_3 = N - 1$		

Individual Feature Analysis: Marijuana Usage in Past 30 Days

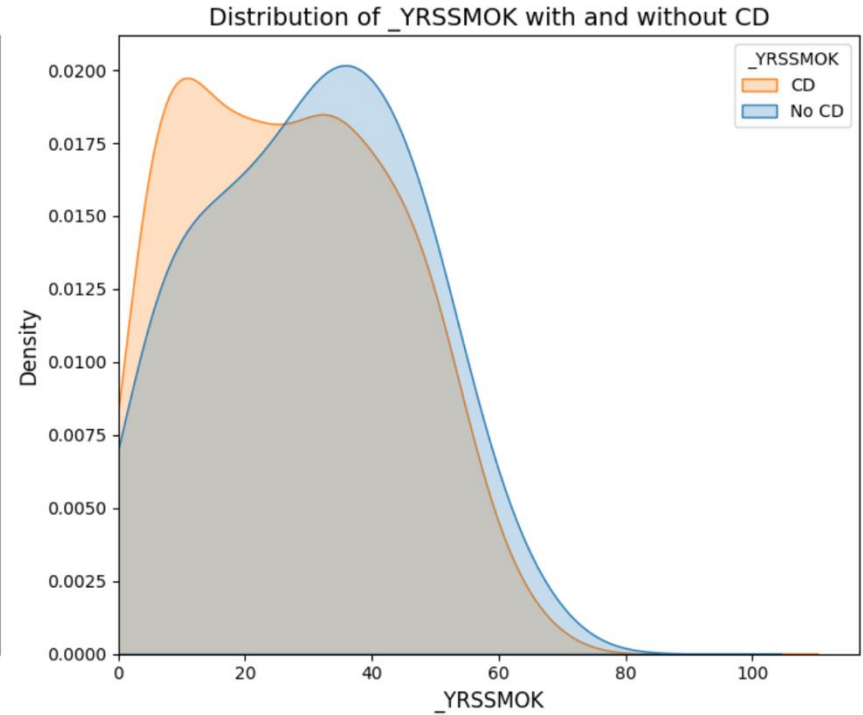
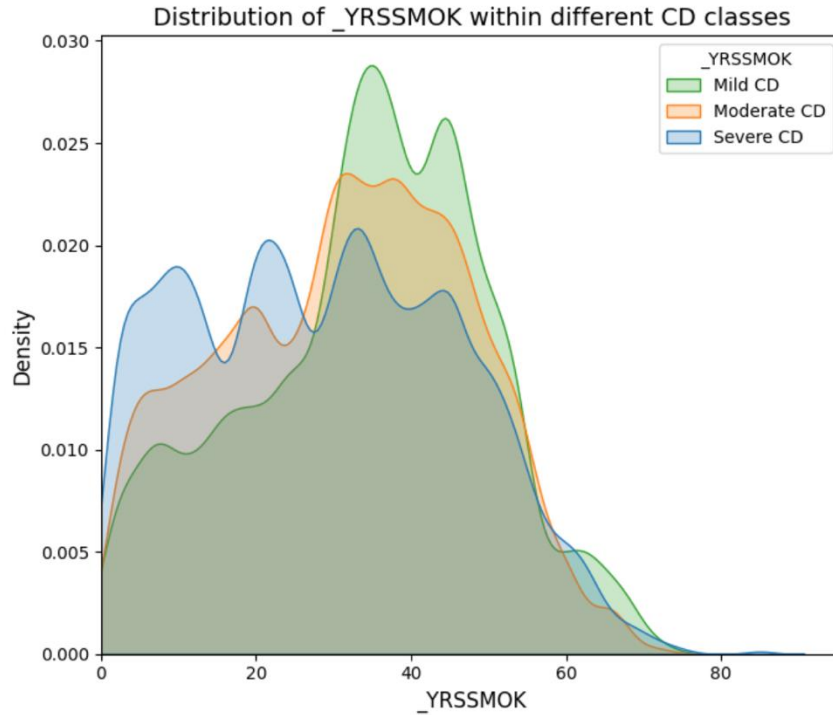
Distribution of MARIJAN1 within different CD classes



Distribution of MARIJAN1 with and without CD



Individual Feature Analysis: Number of years smoked cigarettes



Feature selection (first round)

- For categorical variables
 - Keep those with cramer'V coefficient for multiclass-classfication > 0.1
- For numerical variables
 - Keep those with ANOVA p-value < 0.1
- We also checked the remaining feature one by one and did some adjustment based on factors such as number of missing values, repeated features, etc
- This round of feature selection eliminates features from over 300 to around 60

See Excel

Preprocessing – Conversion

- Data transformations in Ordinal, Binary and Nominal data
 - Binary Conversion :
 - Yes -> 1; No -> 0
 - Ordinal Encoding:
 - For categorical variables with a natural ordered relationship, we used ordered integers to preserve the relationship
 - Nominal One-hot Encoding:
 - For categorical variables where no ordinal relationship exists, we create one new binary variable for each unique level in the variable.

Label: Num Section Na Core Secti Question N Column: 10 Type of Va SAS Variab Question P Question:	Label: Satisfaction with life Section Name: Social Determinants and Health Equity Module Number: 16 Question Number: 1 Column: 361 Type of Variable: Num SAS Variable Name: LSATISFY Question Prologue: Question: In general, how satisfied are you with your life?
Value	Value Label
1	Very satisfied
2	Satisfied
3	Dissatisfied
4	Very dissatisfied
7	Don't know/Not sure
9	Refused
BLANK	Not asked or Missing

Preprocessing – Removal

- Remove “missing” columns after one-hot encoding
 - For nominal variables, new binary variables are created for missing, refused, and not sure responses
 - We remove these columns after applying one-hot encoding.

Label: What is Primary Source of Health Insurance?

Section Name: Health Care Access

Core Section Number: 3

Question Number: 1

Column: 108–109

Type of Variable: Num

SAS Variable Name: PRIMINSR

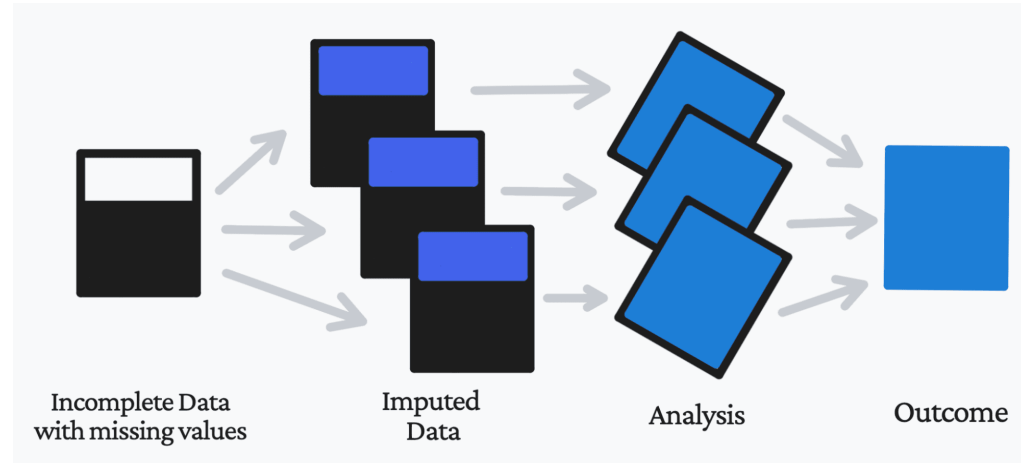
Question Prologue:

Question: What is the current primary source of your health insurance?

Value	Value Label	Frequency
1	A plan purchased through an employer or union (including plans purchased through another person's employer)	161,388
2	A private nongovernmental plan that you or another family member buys on your own	36,931
3	Medicare	135,848
4	Medigap	536
5	Medicaid	29,072
6	Children's Health Insurance Program (CHIP)	188
7	Military related health care: TRICARE (CHAMPUS) / VA health care / CHAMP- VA	15,373
8	Indian Health Service	1,385
9	State sponsored health plan	12,878
10	Other government program	10,630
88	No coverage of any type	23,018
77	Don't know/Not Sure	9,890
99	Refused	7,991
BLANK	Not asked or Missing	4

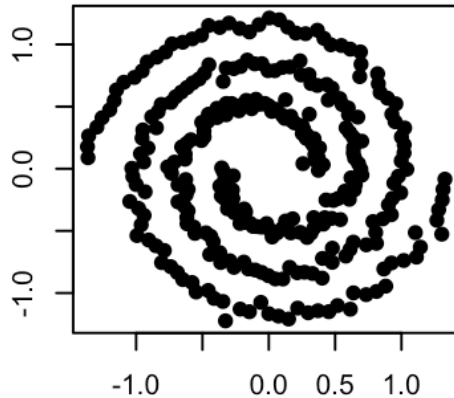
Preprocessing – Imputation

- For features with a reasonable range of missing/refused values, we performed imputation
- Imputation involves replacing missing values with the mean or mode of the observed values in the same feature.
- Numerical - Replace missing values in each feature with the mean
- Categorical - mode of the non-missing values in the same feature.

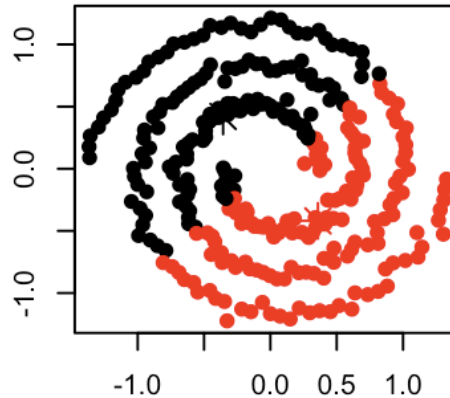


Spectral clustering

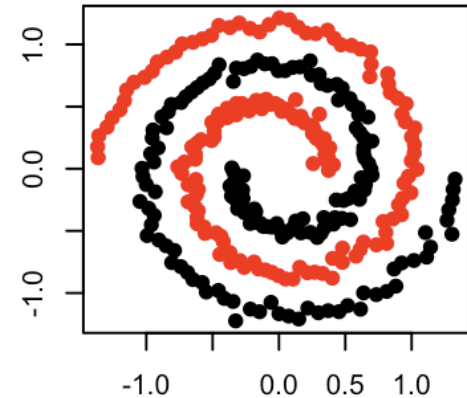
- Spectral clustering treats the data points as nodes in a graph and aims to partition the graph into clusters by analyzing the eigenvalues and eigenvectors of a **similarity matrix** derived from the data



K-means



Spectral clustering

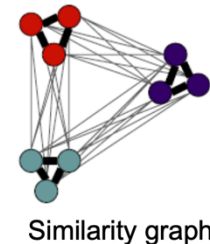
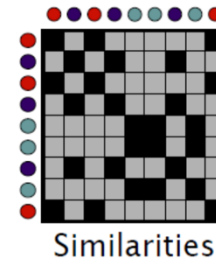
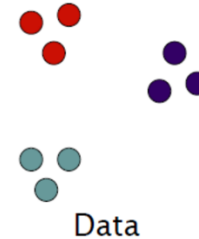


Spectral clustering – Method Review

- Adjacency Matrix A:
 - The row and column indices of an adjacency matrix represent the nodes, and the entries represent the absence or presence of an edge between the nodes.
- Degree Matrix D:
 - The degree of a node is how many edges connect to it. Hence it is simply the sum of the node's row in the adjacency matrix.
- Group Laplacian Matrix L:
 - $L = D - A$
 - L has some nice properties: it is positive semi-definite and the rows sum to zero
- Finding the second smallest eigenvalue is equivalent to solving the optimization:
 - $\lambda_2 = \min_x \frac{x^T L x}{x^T x}$ with respect to the constraint $x^T x = 1$

Spectral clustering – Method Review

- But what is the meaning of $\min_x \frac{x^T L x}{x^T x}$?
- It can be shown that:
 - $\lambda_2 = \min_{\sum x_i = 0} \frac{\sum_{(i,j) \in E} (x_i - x_j)^2}{\sum_i x_i^2}$, where x_i are values we assign to node i
- We also have:
 - $\mathbf{x}$ is a unit vector (by the constraint)
 - $\mathbf{x}$ is orthogonal to $(1, \dots, 1)^T$ (L is positive semi-definite with rows summed up to 0), so that $\sum x_i = 0$
- This means we want to partition the graph in a way such that few edges cross the partition, i.e., the data is split into clusters based on some spectral embedding!



Spectral clustering – Similarity Measure

- Given our dataset is predominantly made up of categorical features, we need to define similarity between two observations
- Overall similarity measure is defined as follows:
 - $S(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^d S(x_i, y_i)$
- We generalized the simple matching measure using the inverse occurrence frequency:

$$S(x_i, y_i) = \begin{cases} 1/p_k(x_i)^2 & \text{if } x_i = y_i \\ 0 & \text{otherwise} \end{cases}$$

- where $p_k(x)$ is the proportion of records in which the k th attribute takes on the value of x in the data set

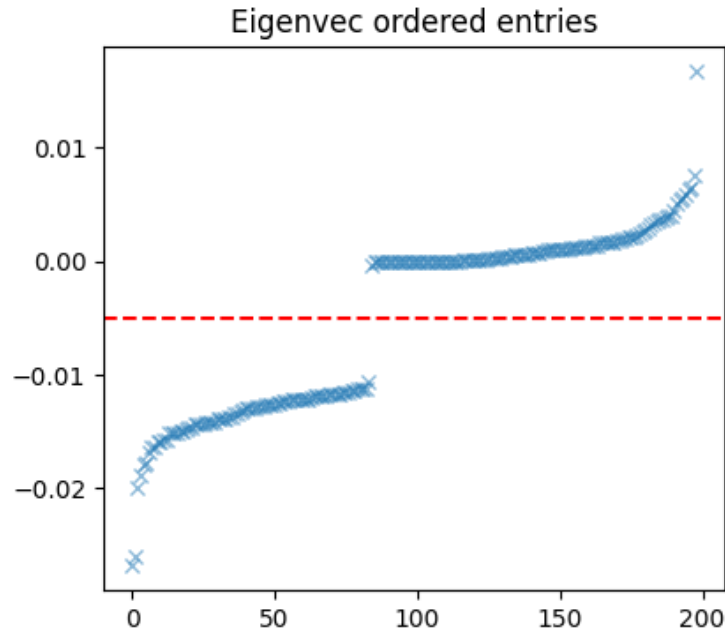
Spectral clustering – Results

- Our data contains over 60k observations, we can only do spectral clustering on subsets of the entire data
- For a random subset of 200 observations, our similarity matrix looks like:

```
array([[1074.551, 163.629, 160.586, ..., 152.133, 152.166, 164.845],  
      [ 163.629, 1441.836, 162.819, ..., 167.135, 385.124, 145.337],  
      [ 160.586, 162.819, 256.626, ..., 192.806, 174.447, 148.314],  
      ...,  
      [ 152.133, 167.135, 192.806, ..., 1183.09 , 151.646, 174.133],  
      [ 152.166, 385.124, 174.447, ..., 151.646, 2309.188, 138.698],  
      [ 164.845, 145.337, 148.314, ..., 174.133, 138.698, 1092.54 ]])
```

- To create the adjacency matrix, we select a threshold of 300 (i.e., we say that two nodes are connected if they have similarity > 300)

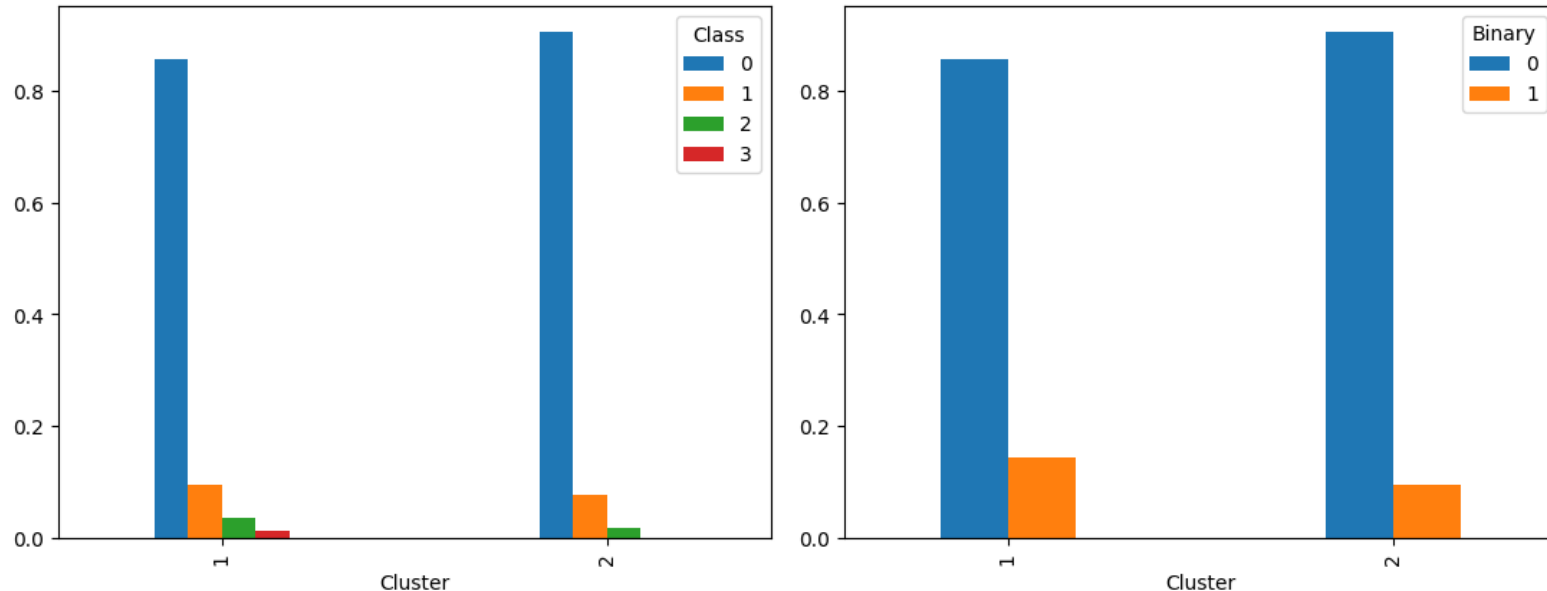
Spectral clustering – Results



- We plot the values of the eigenvector (corresponding to the second smallest eigenvalue)
- We can identify clusters by splitting the sorted vector
- Clearly, we have a split at -0.005 , and the data is split into 2 clusters

Spectral clustering – Why it fails

- Although there exists a clear split of two clusters, the distribution of our target variables are very similar for the two clusters



Feature selection (second round) – Categorical

Class-based Entropy:

$$E(v_i) = - \sum_{j=1}^k p_j \log_2(p_j). \quad E = \sum_{i=1}^r n_i E(v_i) / n.$$

p_j : fraction of data points belonging to the class j for feature v_i .

k : number of target classes

r : number of unique values of the feature

The class-based entropy measure is related to notions of information gain resulting from fixing a specific attribute value

```
Entropy of Very satisfied: 0.5428733
Entropy of DECIDE: 0.5432547
Entropy of _DRNKWK2: 0.5614632
Entropy of _INCOMG1: 0.5651468
Entropy of POORHLTH: 0.5742899
Entropy of MENTHLTH: 0.5761201
Entropy of PHYSHLTH: 0.5761986
Entropy of DIFFALON: 0.5986264
Entropy of Health-Good or Better: 0.5991204
Entropy of Health-Fair or Poor: 0.5992753
Entropy of mental not good:0day: 0.6031132
Entropy of SDHSTRE1: 0.6040705
Entropy of DIFFWALK: 0.6041315
Entropy of ADDEPEV3: 0.6041466
Entropy of mental not good:14+day: 0.6043709
Entropy of Health not good:14+day: 0.6072308
Entropy of WTKG3: 0.6085256
Entropy of _EDUCAG__4.0: 0.6089150
Entropy of SDHISOLT: 0.6089626
Entropy of Health not good:0day: 0.6090801
Entropy of DIFFDRES: 0.6153818
Entropy of SDHBILLS: 0.6235776
Entropy of Dissatisfied: 0.6240143
Entropy of _DRDXAR2: 0.6266938
Entropy of HAVARTH4: 0.6266938
Entropy of BLIND: 0.6267849
```

Feature selection (second round) – Categorical

Gini impurity gain

$$Gini = 1 - \sum_{i=1}^C (p_i)^2$$

$$IG(D_p, x_i) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

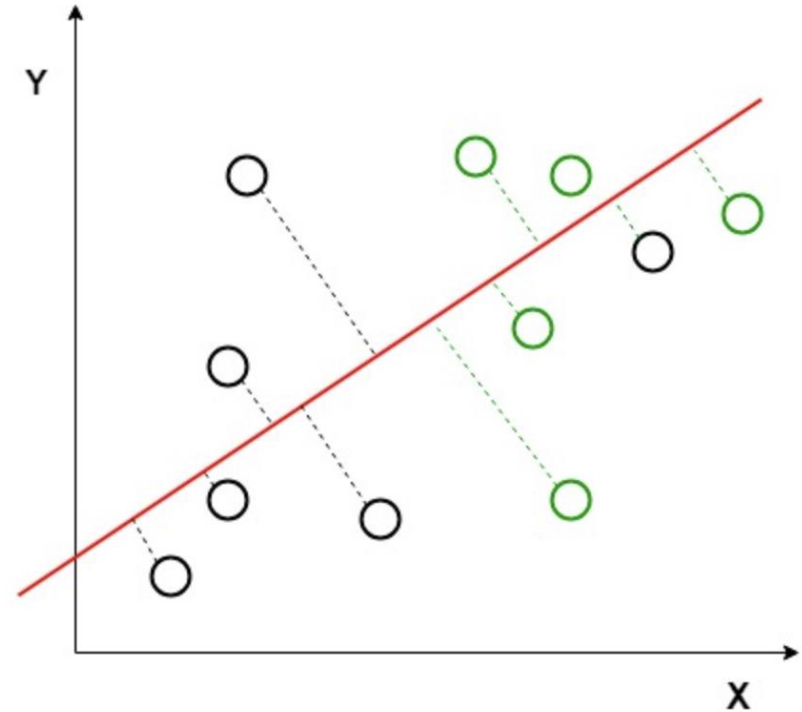
- IG : Information Gain
- x_i : feature to perform the split
- N_p : number of samples in the parent node
- N_{left} : number of samples in the left child node
- N_{right} : number of samples in the right child node
- I : impurity
- D_p : training subset of the parent node
- D_{left} : training subset of the left child node
- D_{right} : training subset of the right child node

Initial Gini Impurity: 0.1973

Gini Impurity Gain for DECIDE: 0.0220
Gini Impurity Gain for POORHLTH: 0.0083
Gini Impurity Gain for MENTHLTH: 0.0081
Gini Impurity Gain for SDHSTRE1: 0.0067
Gini Impurity Gain for PHYSHLTH: 0.0066
Gini Impurity Gain for DIFFALON: 0.0064
Gini Impurity Gain for Health-Good or Better: 0.0063
Gini Impurity Gain for Health-Fair or Poor: 0.0063
Gini Impurity Gain for mental not good:14+day: 0.0061
Gini Impurity Gain for ADDEPEV3: 0.0057
Gini Impurity Gain for DIFFWALK: 0.0055
Gini Impurity Gain for mental not good:0day: 0.0055
Gini Impurity Gain for Health not good:14+day: 0.0051
Gini Impurity Gain for EMPLOY1: 0.0048
Gini Impurity Gain for SDHISOLT: 0.0048
Gini Impurity Gain for Health not good:0day: 0.0041
Gini Impurity Gain for DIFFDRES: 0.0039
Gini Impurity Gain for EMTSUPRT: 0.0038
Gini Impurity Gain for SDHF00D1: 0.0031
Gini Impurity Gain for INCOME3: 0.0029
Gini Impurity Gain for Dissatisfied: 0.0029
Gini Impurity Gain for _INCOMG1: 0.0028
Gini Impurity Gain for SDHBILLS: 0.0025
Gini Impurity Gain for Very satisfied: 0.0023
Gini Impurity Gain for WTKG3: 0.0021

Feature selection (second round) – Numerical

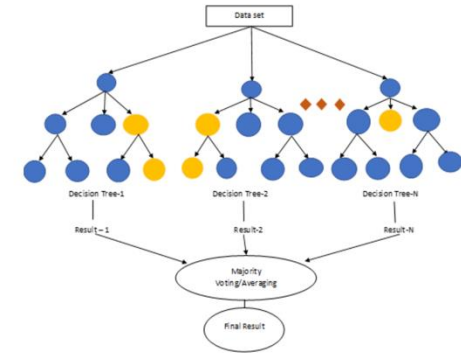
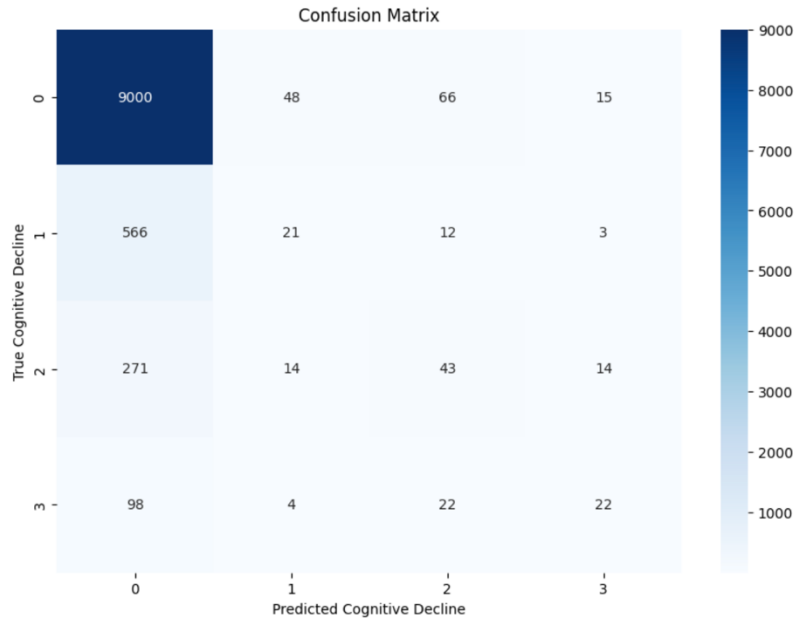
- Definition: Linear Discriminant Analysis (LDA) is a technique used for dimensionality reduction and classification tasks.
- Objective: LDA aims to maximize class separation while minimizing variance within each class.
- Key Features:
 - Supervised learning algorithm
 - Assumes multivariate Gaussian distribution within classes
 - Projects data onto a lower-dimensional space
- We had a total 9 Numerical features and we applied LDA on it.
- After applying we didn't find much of a difference in the accuracy.



The perpendicular distance between the line and points

First Attempt- Random Forest

- Accuracy = 89.4%, seems great but:

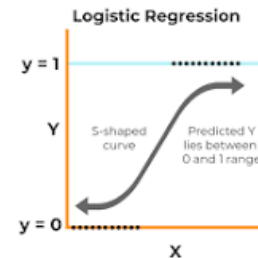


Accuracy: 0.894020941383697

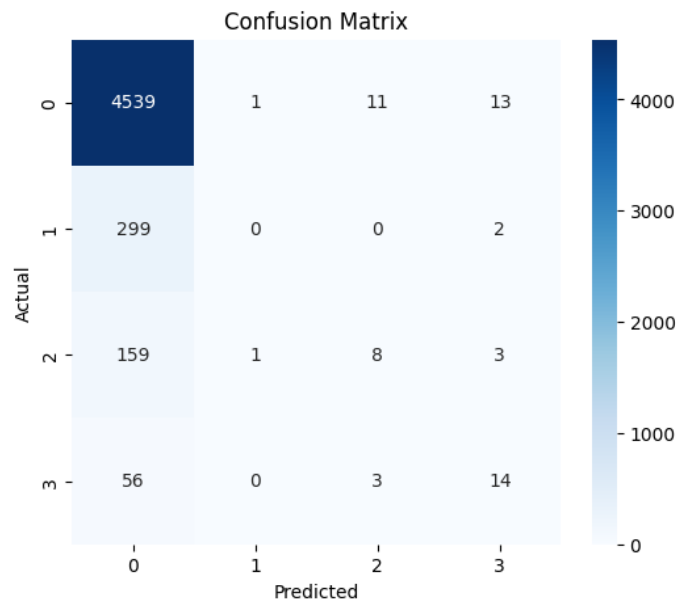
Classification Report:

	precision	recall	f1-score	support
Mild CD	0.00	0.00	0.00	602
Moderate CD	0.52	0.04	0.07	342
No CD	0.90	1.00	0.94	9129
Severe CD	0.39	0.15	0.22	146
accuracy			0.89	10219
macro avg	0.45	0.30	0.31	10219
weighted avg	0.83	0.89	0.85	10219

First Attempt - Logistic Regression



- Same problem:



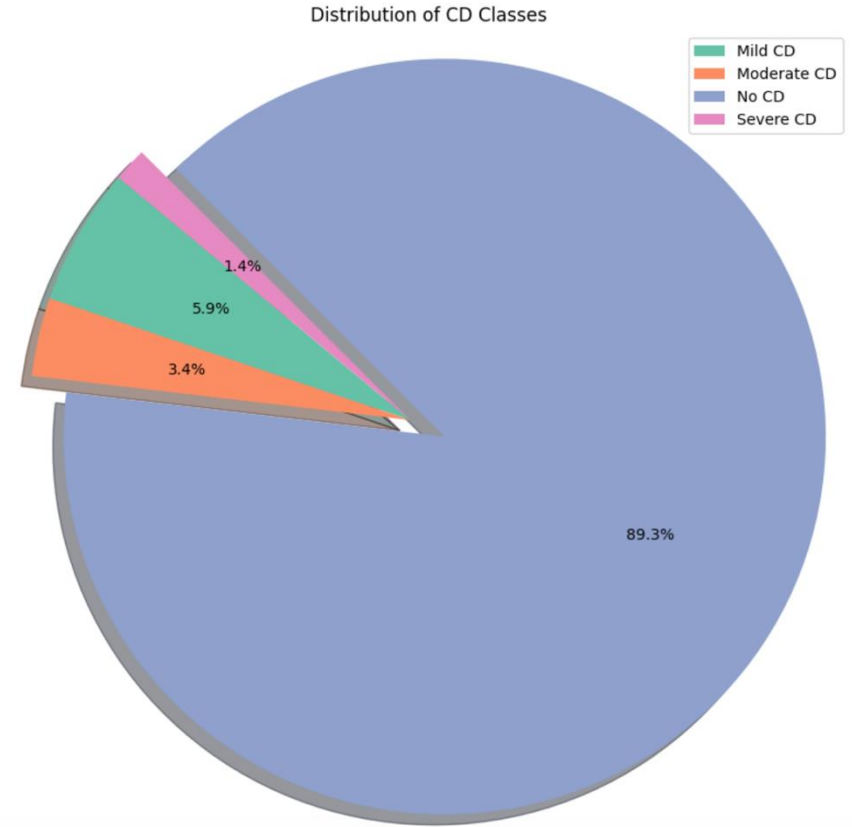
Average Accuracy: 0.8937356700255371

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	4564
1	0.00	0.00	0.00	301
3	0.36	0.05	0.08	171
4	0.44	0.19	0.27	73
accuracy			0.89	5109
macro avg	0.42	0.31	0.32	5109
weighted avg	0.82	0.89	0.85	5109

Imbalance Dataset

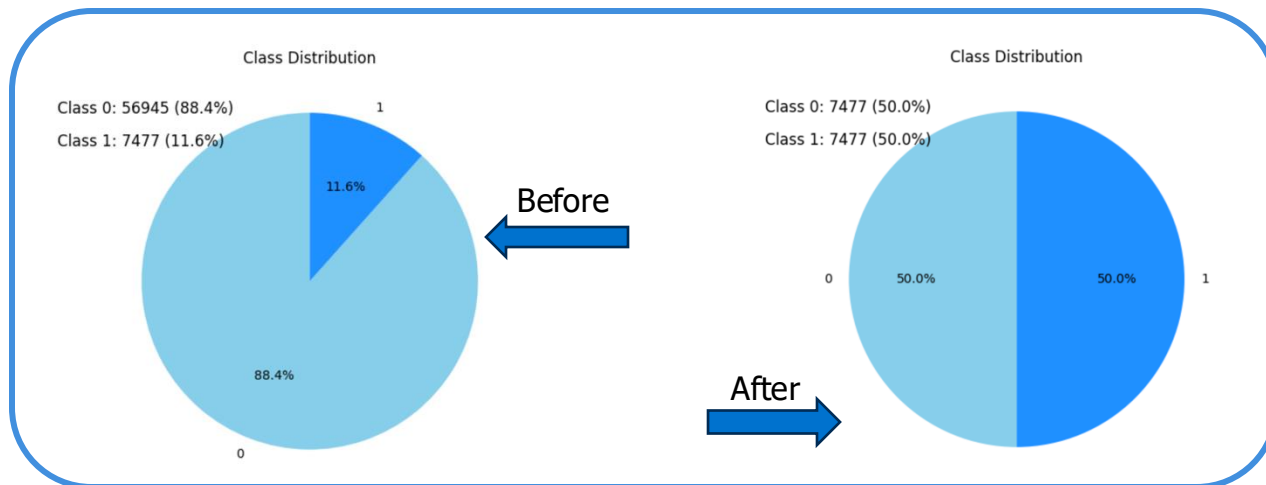
- Imbalanced datasets occur when one class is lower than other.
- Imbalanced datasets are common in various real-world applications.
- **Learn Minority Class:** Minority class patterns may be overshadowed by the majority class, making them difficult to learn.
- **Bias:** Models trained on imbalanced datasets may exhibit bias towards the majority class, leading to poor generalization.
- **Accuracy Paradox:** High overall accuracy may mask poor performance on the minority class.
- **Possible Solutions:** Up-sampling, Down-sampling, SMOTE



Down-sampling

Down-sampling v.s. Up-sampling: Why Down-sampling?

- Faster training
- Reduce overfitting
- Less risk of data leakage
- Improved generalization



Separate majority and minority classes



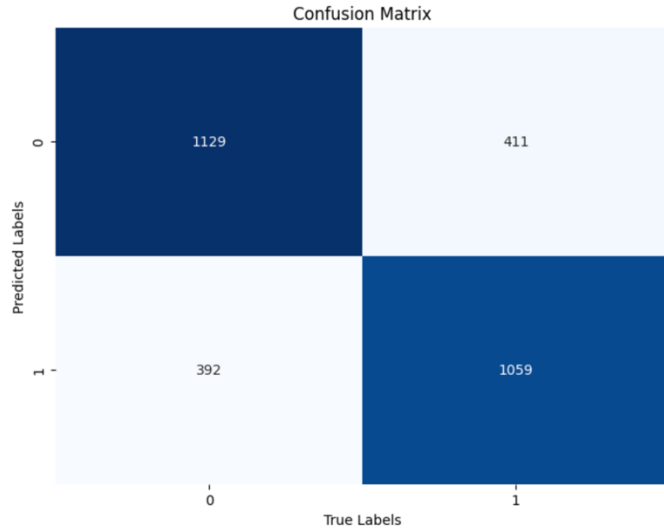
Randomly select samples from majority class without replacement



Shuffle the balanced data

Model accuracy after Down-sampling

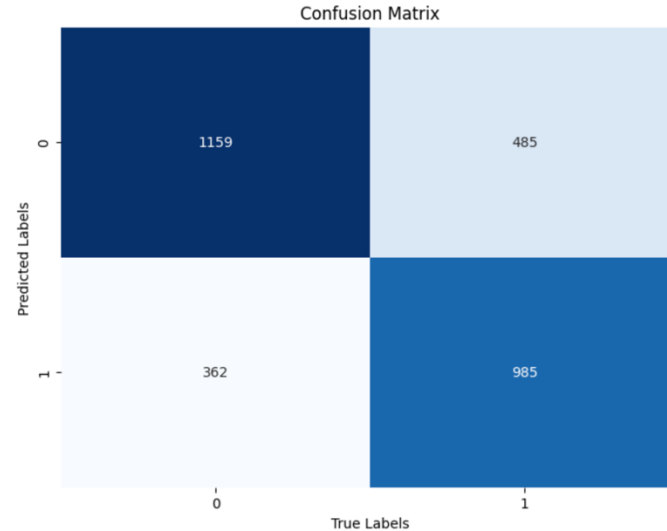
Random Forest (Acc: 73.15%)



Classification Report (Random Forest):

	precision	recall	f1-score	support
Class 0	0.73	0.74	0.74	1521
Class 1	0.73	0.72	0.73	1470
accuracy			0.73	2991
macro avg	0.73	0.73	0.73	2991
weighted avg	0.73	0.73	0.73	2991

Logistic Regression (Acc: 71.68%)



Classification Report (Logistic Regression):

	precision	recall	f1-score	support
Class 0	0.70	0.76	0.73	1521
Class 1	0.73	0.67	0.70	1470
accuracy			0.72	2991
macro avg	0.72	0.72	0.72	2991
weighted avg	0.72	0.72	0.72	2991



Data Augmentation: SMOTE

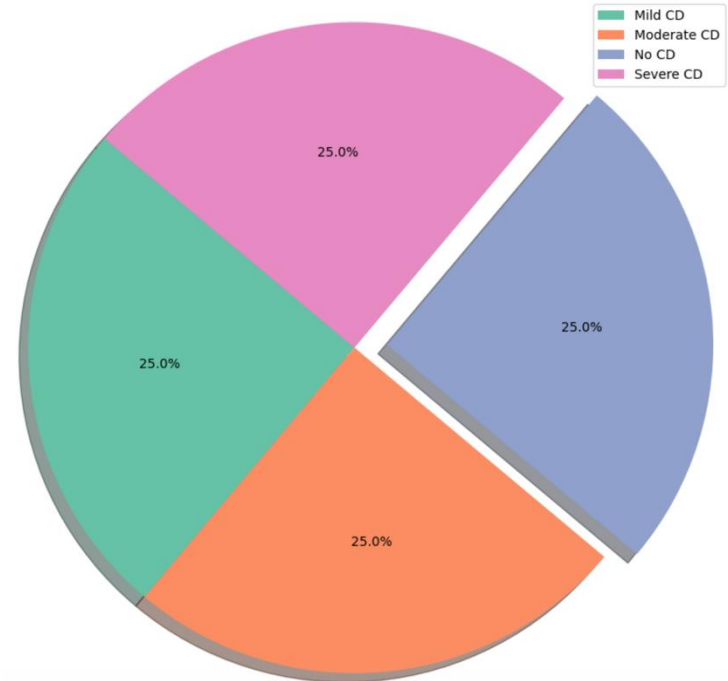


SMOTE: Synthetic Minority
Oversampling Technique



SMOTE - oversampling technique
where the synthetic samples are
generated for the minority class.

Distribution of CD Classes



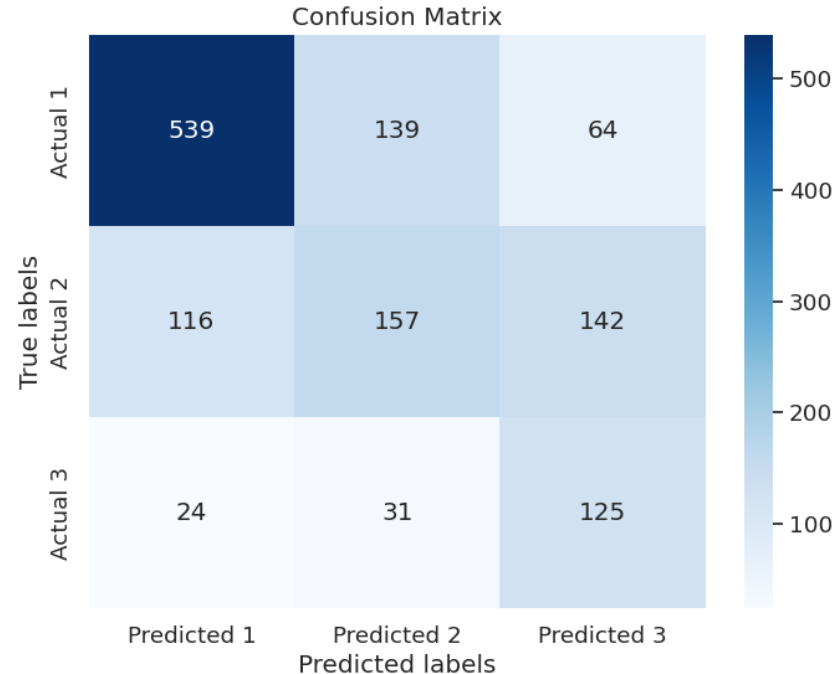
SMOTE – Why it failed



- After applying SMOTE, we got the same accuracy which is 89%.
- **Inadequate Feature Representation:** SMOTE may not improve accuracy if the features used for classification do not effectively capture the underlying patterns in the data.
- **Imbalance Ratio:** If the class imbalance ratio is extremely high or if SMOTE generates synthetic samples that do not accurately represent the underlying distribution of the minority class, it may not lead to improved accuracy.
- **Overfitting:** SMOTE may lead to overfitting if not applied judiciously, especially when generating a large number of synthetic instances.

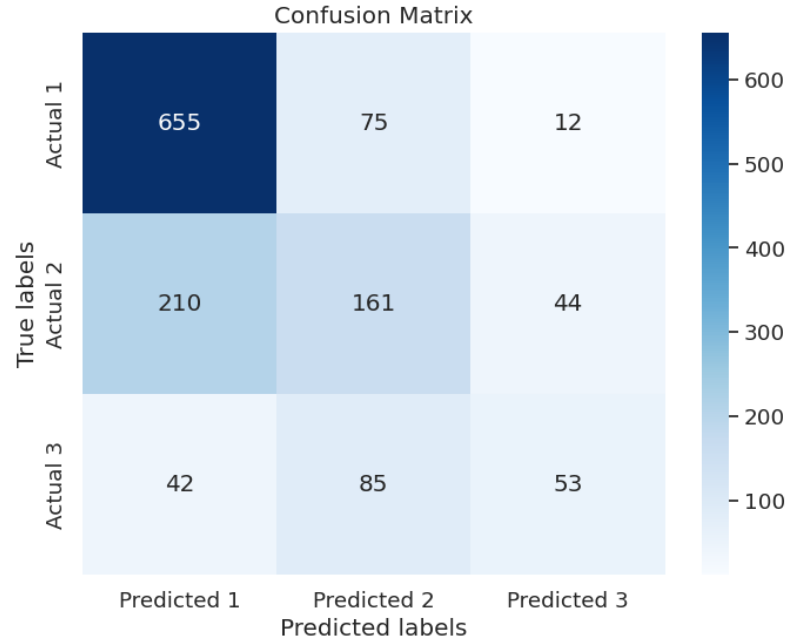
Multi-Class Modelling: Logistic Regression

- We used weights to balance the imbalanced target classes
- We searched through different penalty terms and C values
- The best accuracy we got is 61%



Multi-Class Modelling: Random Forest

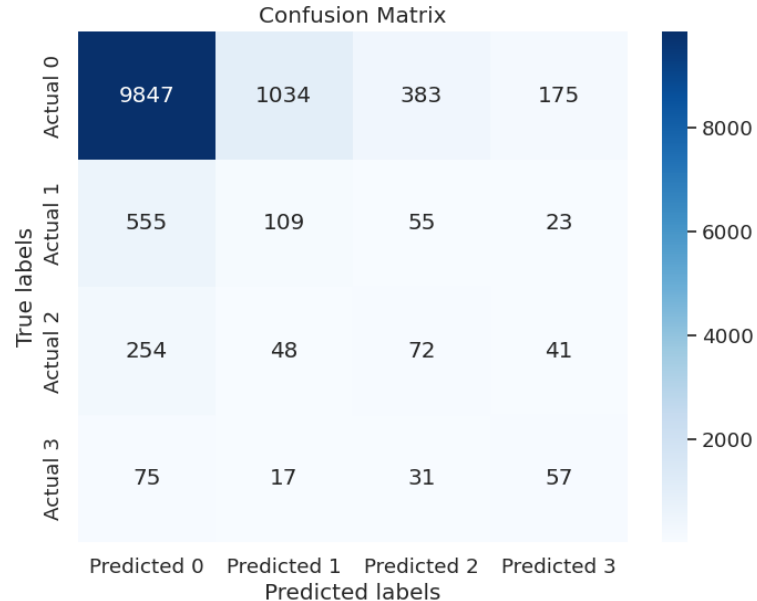
- We used weights to balance the imbalanced target classes
- We searched through different penalty terms and C values
- The best accuracy we got is 65%



Modelling – MLP

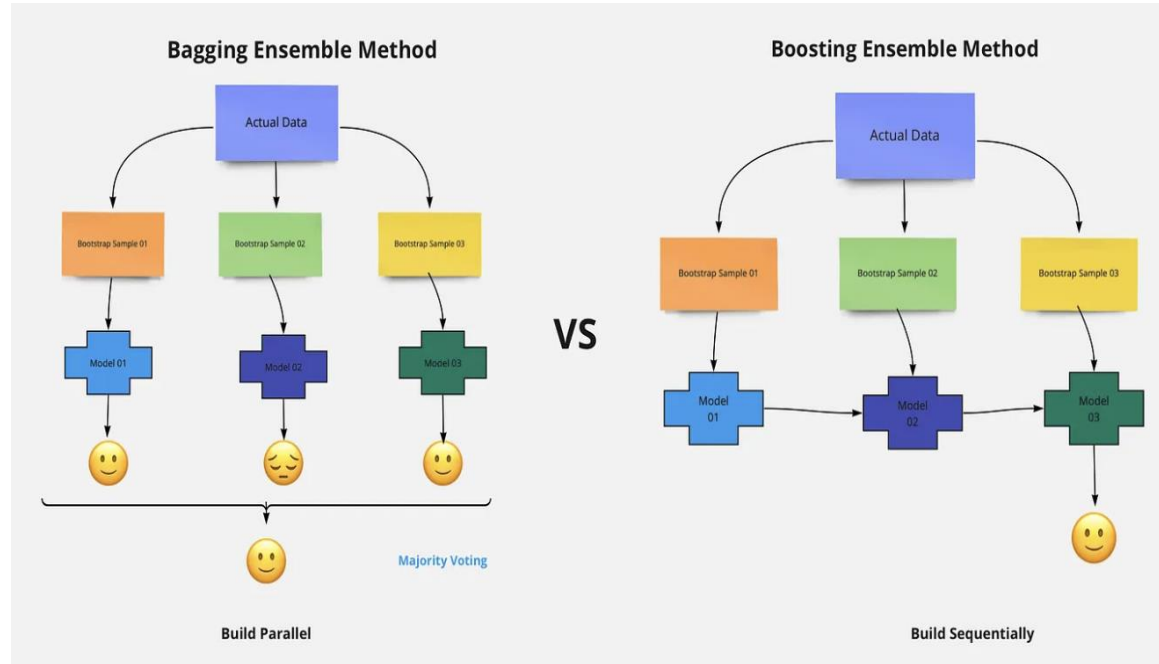
- We trained a multi-layer perceptron model with 4 hidden layers
- Weighted cross-entropy loss function is used
- An accuracy of 78.9% is achieved

Epoch: [1/100]: training accuracy is 0.6701
Epoch: [1/100]: validation accuracy is 0.6360
Epoch: [11/100]: training accuracy is 0.6270
Epoch: [11/100]: validation accuracy is 0.6128
Epoch: [21/100]: training accuracy is 0.6852
Epoch: [21/100]: validation accuracy is 0.6145
Epoch: [31/100]: training accuracy is 0.7466
Epoch: [31/100]: validation accuracy is 0.6739
Epoch: [41/100]: training accuracy is 0.8097
Epoch: [41/100]: validation accuracy is 0.7137
Epoch: [51/100]: training accuracy is 0.8554
Epoch: [51/100]: validation accuracy is 0.7550
Epoch: [61/100]: training accuracy is 0.8878
Epoch: [61/100]: validation accuracy is 0.7291
Epoch: [71/100]: training accuracy is 0.9060
Epoch: [71/100]: validation accuracy is 0.7523
Epoch: [81/100]: training accuracy is 0.9151
Epoch: [81/100]: validation accuracy is 0.7714
Epoch: [91/100]: training accuracy is 0.9438
Epoch: [91/100]: validation accuracy is 0.7980



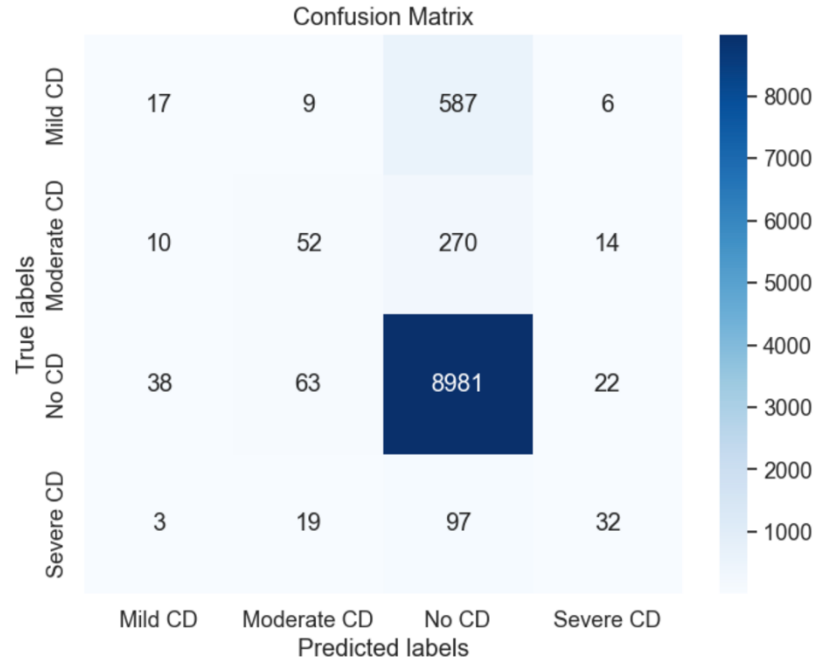
Modelling: XGBoost Classifier

- **Random Forest:** based on decision trees. It builds multiple decision trees during training and outputs the mode of the classes.
- **XGBoost:** based on gradient boosting instead of bagging. It builds an ensemble of decision trees sequentially, where each tree corrects the errors of the previous ones.



Modelling: XGBoost Classifier

- XGBoost is a good option for unbalanced datasets but we cannot trust random forest in these types of cases.
- When the model is encountered with a categorical variable with a different number of classes then there lies a possibility that Random forest may give more preferences to the class with more participation.
- The best XGBoost accuracy we got is 88%



Modelling: Cat Boost



CatBoost

- CatBoost or Categorical Boosting is an open-source boosting library developed by Yandex.
- It is designed for use on problems like regression and classification having a very large number of independent features.
- It's a variant of gradient boosting that can handle both categorical and numerical features.
- Built-in methods for Handling missing values: Unlike other Models, CatBoost can handle missing values in the input data without requiring imputation.
- Built-in cross-validation – CatBoost internally applies a cross-validation method to choose the best hyperparameters for the model.

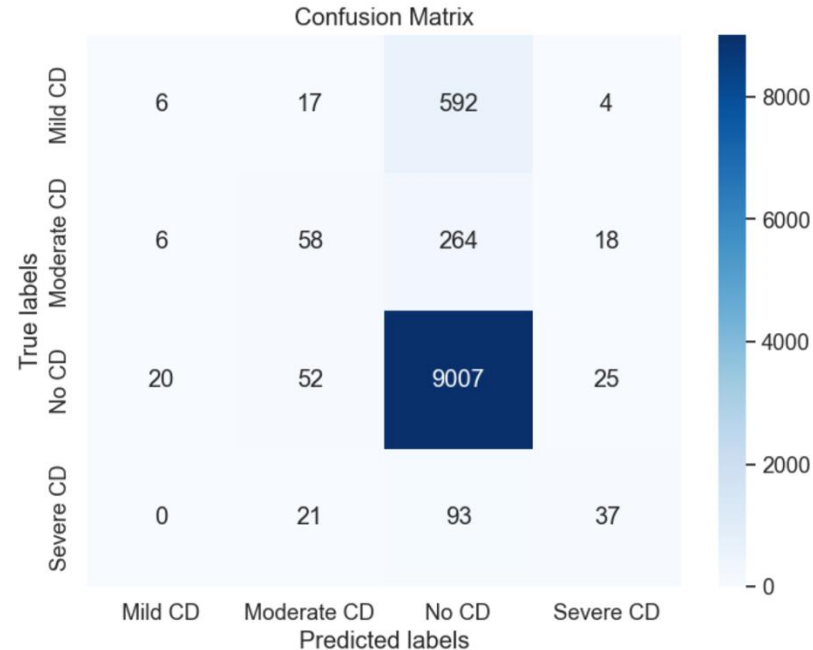
Modelling: Cat Boost



CatBoost

- We got a better accuracy with this technique around 89.11%.

```
988: learn: 0.2165584 total: 1m 1s remaining: 682ms
989: learn: 0.2165145 total: 1m 1s remaining: 620ms
990: learn: 0.2164968 total: 1m 1s remaining: 558ms
991: learn: 0.2164591 total: 1m 1s remaining: 496ms
992: learn: 0.2163879 total: 1m 1s remaining: 434ms
993: learn: 0.2163256 total: 1m 1s remaining: 372ms
994: learn: 0.2162573 total: 1m 1s remaining: 310ms
995: learn: 0.2161768 total: 1m 1s remaining: 248ms
996: learn: 0.2161129 total: 1m 1s remaining: 186ms
997: learn: 0.2159848 total: 1m 2s remaining: 124ms
998: learn: 0.2158981 total: 1m 2s remaining: 62.1ms
999: learn: 0.2158528 total: 1m 2s remaining: 0us
Accuracy: 0.8911937377690803
```



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JOHNS HOPKINS

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